Year 11 Higher

REVISION PACK

Handling Data 4

Name: _____

Class: _____

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: questionnaires

Ali uses this question on a questionnaire to find out how much students spend on snacks. Write down two things that are wrong with the question.

How much do you spend on snacks?
0–1 KD ☐ 1–2 KD ☐ 2–3 KD ☐

Step 1	Is there a time?	No: per day or per week?
Step 2	Do boxes overlap?	Yes: 1 KD is in two boxes
Step 3	Is every answer there?	No box for more than 3 KD

Practice: questionnaires

1.1 Huda uses this question on a questionnaire to find out how much time students spend on homework. Write down two things that are wrong with the question.

How many hours of homework do you do?
0–2 ☐ 2–4 ☐ 4–6 ☐

1.2 Majed uses this question on a questionnaire to find out how often people exercise. Write down two things that are wrong with the question.

How many times do you exercise?
1–3 ☐ 3–5 ☐ 5–7 ☐

1.3 Layla uses this question on a questionnaire to find out how much people spend on lunch. Write down two things that are wrong with the question.

How much do you spend on lunch?
0–2 KD ☐ 2–4 KD ☐ 4–6 KD ☐

Practice: questionnaires (continued)

1.4 Omar uses this question on a questionnaire to find out how much time people spend on their phones. Write down two things that are wrong with the question.

How many hours do you spend on your phone?
0–2 ☐ 2–4 ☐ 4–6 ☐

1.5 Mona uses this question on a questionnaire to find out how much people spend on clothes. Write down two things that are wrong with the question.

How much do you spend on clothes?
0–10 KD ☐ 10–20 KD ☐ 20–30 KD ☐

1.6 Fahad uses this question on a questionnaire to find out how many books students read. Write down two things that are wrong with the question.

How many books do you read?
1–2 ☐ 2–3 ☐ 3–4 ☐

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: averages from a frequency table

The table shows the numbers of goals scored by a team in 20 matches.

Number of goals	0	1	2	3
Frequency	5	8	4	3

(a) Write down the mode. (b) Work out the mean number of goals.

(a)	Step 1	Highest frequency	8
	Step 2	Mode is that value	1
(b)	Step 1	Goals × frequency	$0 + 8 + 8 + 9 = 25$
	Step 2	Total frequency	$5 + 8 + 4 + 3 = 20$
	Step 3	Mean = total ÷ frequency	$25 \div 20 = 1.25$

Practice: averages from a frequency table

1.7 The table shows the numbers of goals scored by a team in 25 matches.

Number of goals	0	1	2	3	4
Frequency	3	7	8	5	2

(a) Write down the mode. (b) Work out the mean.

1.8 The table shows the numbers of emails Sara got on 25 days.

Number of emails	1	2	3	4	5
Frequency	2	9	6	4	4

(a) Write down the mode. (b) Work out the mean.

1.9 The table shows the numbers of visitors to a shop each hour for 25 hours.

Number of visitors	0	1	2	3	4
Frequency	6	5	4	8	2

(a) Write down the mode. (b) Work out the mean.

1.10 The table shows the numbers of pets owned by 20 students.

Number of pets	0	1	2	3	4
Frequency	5	9	4	1	1

(a) Write down the mode. (b) Work out the mean.

1.11 The table shows the numbers of books read by 20 students.

Number of books	0	1	2	3	4
Frequency	2	4	8	4	2

(a) Write down the mode. (b) Work out the mean.

1.12 The table shows the numbers of brothers and sisters of 20 students.

Number of brothers and sisters	0	1	2	3	4
Frequency	3	8	6	2	1

(a) Write down the mode. (b) Work out the mean.

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: a missing number from the mean

The mean of four numbers is 9. Three of the numbers are 7, 8 and 12. Work out the fourth number.

Step 1	Mean × how many	$4 \times 9 = 36$
Step 2	Add the three	$7 + 8 + 12 = 27$
Step 3	Total − the three	$36 - 27 = 9$

Practice: a missing number from the mean

1.13 The mean of five numbers is 16. Four of the numbers are 12, 14, 19 and 20. Work out the fifth number.

1.14 The mean of six numbers is 11. Five of the numbers are 8, 9, 13, 14 and 10. Work out the sixth number.

1.15 The mean of five numbers is 23. Four of the numbers are 18, 25, 27 and 20. Work out the fifth number.

1.16 The mean of five numbers is 12. Four of the numbers are 9, 11, 14 and 15. Work out the fifth number.

1.17 The mean of four numbers is 20. Three of the numbers are 18, 25 and 16. Work out the fourth number.

1.18 The mean of six numbers is 15. Five of the numbers are 12, 18, 14, 20 and 11. Work out the sixth number.

Handling Data 4 · Pie charts (Lesson 4.3)

Worked example: pie chart angles

In a survey, 60 people were asked to name their favourite colour. 21 people chose blue. Work out the angle of the sector for blue.

Step 1	Angle for one person	$360 \div 60 = 6^\circ$
Step 2	Multiply by the number who chose blue	$21 \times 6 = 126$
Step 3	Answer	126°

Practice: pie chart angles

2.1 90 people were asked to name their favourite sport. 25 chose football. Work out the angle of the sector for football.

2.2 120 students were asked to name their favourite subject. 38 chose maths. Work out the angle of the sector for maths.

2.3 40 people were asked what they drink at breakfast. 13 chose tea. Work out the angle of the sector for tea.

2.4 A car park has 72 cars. 20 of the cars are red. Work out the angle of the sector for red cars in a pie chart.

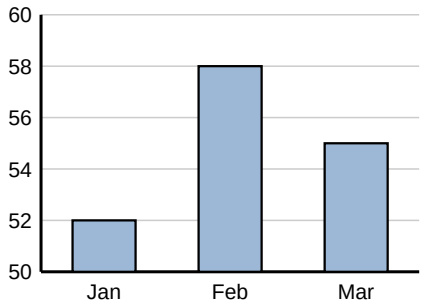
2.5 A pie chart shows how 30 students travel to school. The sector for walking has an angle of 84°. How many students walk to school?

2.6 A pie chart shows the favourite food of 180 people. The sector for rice has an angle of 130°. How many people chose rice?

Handling Data 4 · Misleading charts (Lesson 4.4)

Worked example: what is misleading?

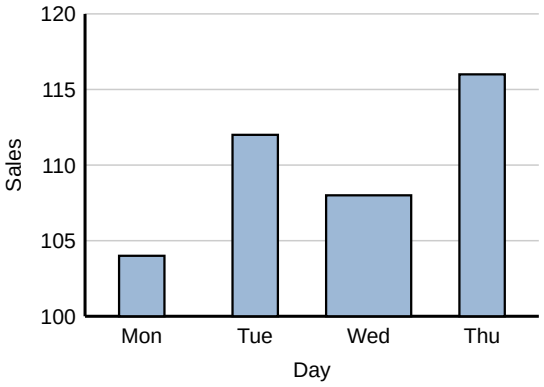
The bar chart shows the number of cars sold by a garage in three months. Write down two things that are misleading about this bar chart.



Step 1	Look at the vertical axis	It starts at 50, not 0, so the differences look bigger
Step 2	Look at the labels	There is no label on the vertical axis and no title
Step 3	Answer	Axis does not start at 0; no axis label or title

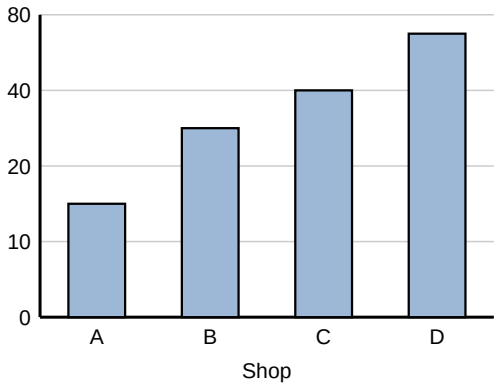
Practice: what is misleading?

2.7 The bar chart shows the sales of a shop on four days. Write down two things that are misleading about this bar chart.



Practice: what is misleading? (continued)

2.8 The bar chart shows the number of customers at four shops. Write down two things that are misleading about this bar chart.



2.9 The bar chart shows the number of members of a club in three years. Write down two things that are misleading about this bar chart.

